

YALU CAI

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Education

Rutgers University

PhD in Computer Science, advised by Prof. Santosh Nagarakatte

Piscataway, NJ, USA

Sept. 2025 – Present

Duke University

Master of Science in Computer Science

Durham, NC, USA

Aug. 2020 – May 2022

- GPA: 3.837/4.0

Cornell University

Bachelor of Science in Biological Engineering

Ithaca, NY, USA

Aug. 2018 – May 2020

- GPA: 3.898/4.3

Zhejiang University

Bachelor of Engineering in Biological Engineering

Hangzhou, China

Sept. 2016 – Jul. 2018

- GPA: 3.77/4.0
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Publications

Proof-Carrying-BPF Programs for the eBPF Verifier

Yalu Cai, Santosh Nagarakatte

eBPF '26: The 4th Workshop on eBPF and Kernel Extensions, Prague, Czech Republic, Sept. 2026

Research

Alivio: Proof-Carrying BPF Programs | Rust, CVC5

Sept. 2025 – Present

- A userspace mirror of the Linux kernel's eBPF verifier that generates machine-checkable SMT proofs for the program points the kernel's abstract domains cannot prove safe, widening the class of accepted programs without growing the kernel's trusted computing base.
- Matches the kernel verifier's verdict on all 3,267 BPF selftests; its proof bundles let a patched kernel load 189 production programs from Cilium, Calico, and Inspektor Gadget that the stock verifier rejects, with $3.4\times$ faster loads than prior work.

Verified Translator from EVM Bytecode to eWASM | Coq

Sept. 2021 – May 2022

- Implements a translator that translates a subset of Ethereum Virtual Machine Bytecode program into WebAssembly.
- Verifies the translator for Semantic Preservation using Coq.

Typechecker for ssm-edsl | Haskell

May. 2021 – Oct. 2021

- Worked with the ssm-edsl group from Columbia University and Chalmers University (Sweden) to implement a typechecker for SSM (Sparse Synchronous Model), a language designed to aid real-time data collection.
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Work Experience

Software Engineer

Jun. 2022 – Aug. 2025

CertiK LLC

New York, NY, USA

- Designed and developed compilers from Solidity to Dafny and Boogie. Especially involved in the design of the intermediate representation language Swivel, which connects Solidity to both Dafny and Boogie.
- Extended Solidity with a specification language which is adapted from Java Modeling Language (JML).
- Performed deductive verification on Solidity projects translated into Dafny.
- Developed a set of Python scripts that automates the building and running of the company's formal verification tool on Solidity projects, greatly decreasing development and testing time.

Graduate Teaching Assistant

Aug. 2020 – May 2022

Duke University

Durham, NC, USA

- Teaching assistant for CS310 Operating Systems; CS230 Discrete Math; CS334 Mathematical Foundations of CS
- Responsible for organizing recitation sessions, holding office hours, and grading homework and exams.

Coding Projects

EVM Bytecode Compiler | *LLVM, Python*

April 2024 – Now

- A compiler of EVM bytecode to machine code using LLVM (llvmlite).
- The compiler supports basic opcodes including STOP, LOAD, STORE, SUB, ADD, PUSH, POP, MSTORE, MLOAD, etc.

Tiger Compiler | *Standard ML*

Jan. 2021 – May 2021

- A compiler for Tiger, a small language with nested functions, record values with implicit pointers, arrays, integer and string variables, and a few simple structured control constructs.
- Features: type checking and bounds checking for array subscriptions.

Paxos Distributed System | *Java*

Jan. 2022 – May 2022

- A Java implementation of the famous Paxos protocol, which utilizes state machine replication to tolerate server failures in a distributed system.
- This project models the clients, servers, messages and timers in a distributed system. It also implements algorithms for stable leader election and garbage collection.

Movie Matcher | *Swift*

Aug. 2021 – Dec 2021

- Movie Matcher is an iOS app that allows users to like or dislike a movie by swiping to the left or right; it will recommend new movies to users based on their preferences.
- Developed a data persistence mechanism that works by saving data onto files or databases.

RISC-V CPU | *Logism*

Oct. 2020 – Dec. 2020

- Implemented a **pipelined** RISC-V CPU in the circuit simulation software Logism. The CPU supports 33 basic operations with arithmetic, logic, memory manipulation and control flow functionalities.

Technical Skills

Languages: Rust, C/C++, OCaml, Python, Coq, Haskell, Golang, Java, SQL, Swift, JavaScript, TypeScript

Developer Tools: Git, Linux, Docker, LLVM, SMT solvers (CVC5)